

Pure-State Pi-Base Does Not Force a Separable Right-Support Net

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Source Problem and Lane Context

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, it asks for a noncommutative C^* -algebra characterization:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

The previous lane-19 packet

`2311.14257_separable_right_support_net_ubcp`

proved the following useful sufficient condition: if A has positive contractions e_n such that $\overline{Ae_n}$ is separable for every n and

$$\|a\| = \sup_n \|ae_n\| \quad (a \in A),$$

then A has UBCP. The present packet shows that pure-state countable π -base cannot force this support-net condition, even in the commutative case.

The Counterexample

Let

$$K = ([0, 1] \times \{0, 1\}) \setminus \{(0, 0), (1, 1)\}$$

with the lexicographic order:

$$(s, i) < (t, j) \iff s < t, \text{ or } s = t \text{ and } i < j.$$

Equip K with the order topology. This is the standard endpoint-free split interval, also called the double-arrow compactum. Let $A = C(K)$.

Theorem 1. *The commutative C^* -algebra $A = C(K)$ has the following properties:*

1. *its pure state space $P(A) \cong K$ has a countable π -base;*
2. *A has UBCP;*
3. *no nonzero positive $e \in A$ has separable closed right ideal $\overline{Ae}$.*

Consequently, pure-state countable π -base does not imply the existence of a separable right-support net.

Idea of the Proof

The endpoint-free split interval is small from the point of view of π -bases: rational order intervals form a countable π -base. But it is locally large from the point of view of separability of function algebras: every nonempty open subset contains a copy of a Sorgenfrey interval, hence is not second countable.

For $C(K)$, the closed principal ideal generated by e is

$$\overline{C(K)e} = \{f \in C(K) : f = 0 \text{ on } Z(e)\} \cong C_0(\text{coz}(e)).$$

If this ideal were separable, then $\text{coz}(e)$ would be second countable. That is impossible for nonzero e , because $\text{coz}(e)$ is a nonempty open subset of the split interval.

Proof

Lemma 1. *The split interval K has a countable π -base.*

Proof. For rational $0 \leq p < q \leq 1$, include the order intervals

$$((p, 1), (q, 0)).$$

Also include the rational endpoint rays, where the minimum of K is $(0, 1)$ and the maximum is $(1, 0)$,

$$[(0, 1), (q, 0)) \quad (0 < q \in \mathbb{Q}) \quad \text{and} \quad ((p, 1), (1, 0)] \quad (p \in \mathbb{Q}, p < 1).$$

This family is countable.

Let $W \subset K$ be nonempty open. By the order-topology basis, W contains a nonempty order interval or endpoint ray. Such a set has positive width in the first coordinate: otherwise it would be contained in the empty gap between an adjacent pair $(t, 0) < (t, 1)$. Choose rationals $p < q$ so that the corresponding rational interval, or one of the endpoint rays, lies inside that basic open set. Hence some member of the displayed countable family is contained in W , proving that the family is a π -base. $\square$

Lemma 2. *No nonempty open subset of K is second countable.*

Proof. Let $W \subset K$ be nonempty open. By Lemma 1, W contains an interval

$$((p, 1), (q, 0))$$

with $p < q$. Inside this interval, consider the upper copy

$$S_{p,q} = \{(t, 1) : p < t < q\}.$$

With the subspace topology inherited from K , this is a Sorgenfrey interval: a neighborhood base at $(t, 1)$ is given by sets corresponding to

$$[t, s) \quad (t < s < q).$$

A nontrivial Sorgenfrey interval is not second countable. Indeed, if $\mathcal{B}$ were a countable base, then for each $t \in (p, q)$ one could choose $B_t \in \mathcal{B}$ such that

$$t \in B_t \subset [t, q).$$

If $s < t$, then B_t cannot equal B_s , because $B_t \subset [t, q)$ does not contain s , while B_s contains s . Thus $\mathcal{B}$ would contain uncountably many distinct members, a contradiction.

Therefore $S_{p,q}$ is not second countable. Since subspaces of second countable spaces are second countable, W itself is not second countable. $\square$

Lemma 3. *If $0 \neq e \in C(K)_+$, then $\overline{C(K)e}$ is not separable.*

Proof. Let

$$U = \text{coz}(e) = \{x \in K : e(x) \neq 0\}.$$

This is a nonempty open subset of K . The closed ideal generated by e is

$$\overline{C(K)e} = \{f \in C(K) : f|_{K \setminus U} = 0\},$$

which is canonically isometric to $C_0(U)$.

For a locally compact Hausdorff space U , $C_0(U)$ is separable only if U is second countable. One way to see this is to take a countable dense set in $C_0(U)$; the cozero sets of its rational level cuts form a countable base for U . But Lemma 2 says that no nonempty open subset of K , including U , is second countable. Hence $\overline{C(K)e} \cong C_0(U)$ is not separable. $\square$

Proof of Theorem 1. Since $A = C(K)$ is commutative and unital, its pure state space is homeomorphic to K , via point evaluations. Lemma 1 gives a countable π -base for $P(A)$.

The source paper's commutative theorem gives UBCP for $C(K)$, because K has a countable π -base.

Finally, Lemma 3 says that no nonzero positive $e \in A$ has separable $\overline{Ae}$. Therefore a separable right-support net cannot exist: every element of such a net would have to be zero, and then $\sup_n \|1 \cdot e_n\| = 0 \neq 1$. $\square$

Consequence for the C*-BCP Program

This counterexample does not threaten the source theorem. In fact, it is a commutative example where the source theorem works perfectly:

$$K \text{ has a countable } \pi\text{-base} \implies C(K) \text{ has UBCP.}$$

What fails is the lane-19 intermediate hope that pure-state countable π -base should imply a countable separable right-support net.

Thus the right-support-net theorem remains a strong sufficient condition, but it is not the necessary bridge to the full classification. A successful noncommutative pure-state proof must imitate the commutative scalar bump mechanism more directly. Equivalently, it needs a weaker local coefficient or rank-one bump structure, not separability of the whole principal right ideal $\overline{Ae}$.

Novelty and Search Bounds

Local indexes were searched for arXiv:2311.14257, pure state π -base, right-support net, split interval, double arrow, separable principal ideal, and commutative BCP. The split interval is a standard compact ordered space, but no previous run packet recorded this obstruction to the right-support-net bridge. The packet is therefore a counterexample to an agent-generated intermediate lemma, not a claim of a new counterexample to the published source question.

Human Review Focus

Reviewers should check:

1. the countable π -base for the split interval;
2. the identification of the upper copy with a Sorgenfrey interval;
3. the proof that every nonempty open subset of K is not second countable;
4. the standard commutative ideal identity $\overline{C(K)}e \cong C_0(\text{coz}(e))$;
5. the distinction between refuting the support-net bridge and refuting the source problem.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.
- [3] Run fa_banach_001, lane 19, *Separable Right-Support Nets Force UBCP*, solution packet 2311.14257_separable_right_support_net_ubcp, 2026.